

The image processing protocol of “Comparison of an algorithm quantitatively estimating epifascial fibrosis in three-dimensional computed tomography images to other clinical lymphedema grading methods”

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1. The original CT image file (Fig. P1) is loaded in the MATLAB*.

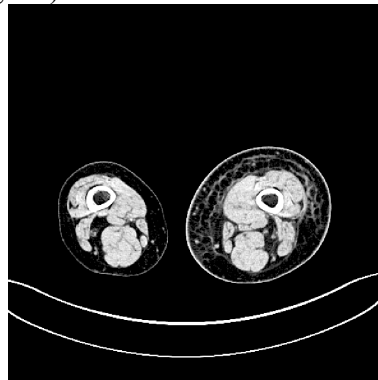


Fig. P1. The original CT image.

2. The reticulated limb region is selected (Fig. P2).

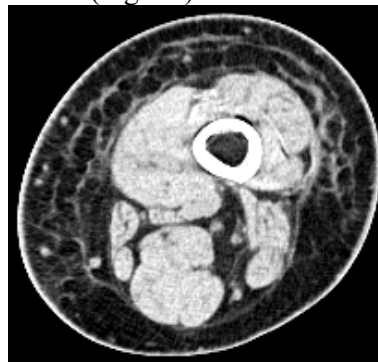


Fig. P2. The reticulated limb region (the right part of Fig. P1).

3. In order to select only skin, muscle, and bone, pixels from -50 HU to 26 HU are eliminated. Any remained noise is removed by a simple opening filter or just manually. (Fig. P3).



Fig. P3. The remained skin, muscle, and bone.

4. The connected regions are discriminated using **regionprops** function in the MATLAB*. After that, under-20-pixels regions are eliminated.



Fig. P4. The recognized connected regions over 20 pixels.

5. The recognized image was converted into the binary image and then the outmost skin is extracted using **bwboundaries** function in the MATLAB*. The extracted skin is dilated to have thick boundary.

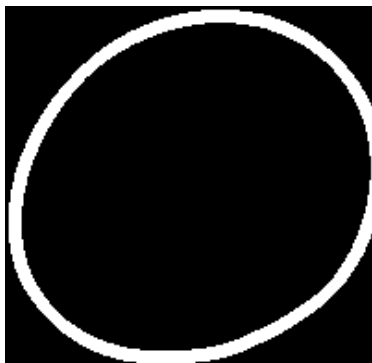


Fig. P5. The thick outmost boundary.

6. Using the thick outmost boundary (Fig. P5) the outmost skin is eliminated from the connected regions (Fig. P4).



Fig. P6. The connected regions inside only.

7. The inside regions are reconstruction as one connected region using **imclose** function in the MATLAB*.

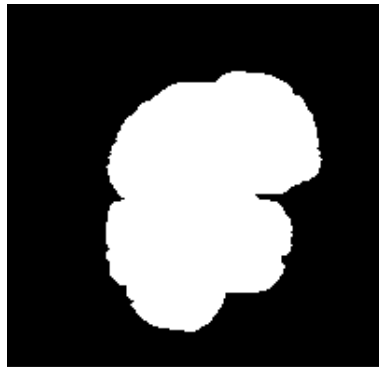


Fig. P7. The reconstructed inside region.

8. The thick outmost boundary (Fig. P5) is added to the reconstructed inside region (Fig. P7) by logical operations.



Fig. P8. The thick outmost boundary and the reconstructed inside region.

9. The black and white pixels of the merged regions (Fig. P8) are reversely converted.



Fig. P9. The reversely converted boundary and inside.

10. Using the reversely converted region (Fig. P9), the reticulated patterns are extracted as a donut-like shape from the reticulated limb region (Fig. P2) by logical operations.

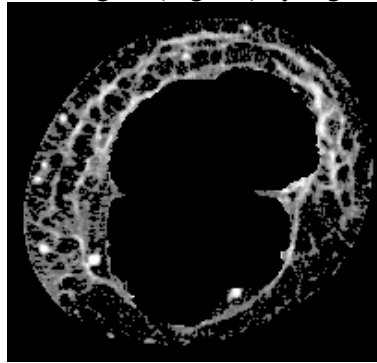


Fig. P10. The extracted reticulated patterns.

11. The nonblack pixels of the reticulated patterns (Fig. P10) are regarded as the epifascial lymph-proliferative fibrosis.

* MATLAB is the mathematical package of MathWorks Inc., Natick, MA, U.S.A.